DATE

Bonus Simulation

~~So~~ As all the 4 MOSFET's used are identical (due to mirroring), all will have same value for ref current that is 5 mA.

$$10 = \frac{1}{2} \times \overset{K_n}{\cancel{20mA}} \times (V_{GS} - V_T)^2$$

$$20 = 20mA \times (V_{GS} - V_T)^2$$

$$\cancel{\frac{20}{20} \times 1000 = (V_{GS} - V_T)^2}$$

$$V_{GS} \Rightarrow \pm 0.7 + 1$$

$$V_{GS} \Rightarrow \begin{cases} \rightarrow 1.7 \text{ [we use this]} \\ \rightarrow 0.3 \text{ [as if use this } V_{GS} < V_T] \end{cases}$$

$$V_{GS} = 1.7 V$$

$$V_{GS} = V_{DS} \Rightarrow 1.7 V$$

$$1.7 = V_0 - (-15)$$

$$V_{D_1} = -13.3 V$$

$$V_{S_3} = V_{D_1} = -13.3 V$$

So Now in M_3: Similar to above

$$V_{GS3} = 1.7 V$$

DATE

$$V_{G7} - (-13.3) = 1.7$$

$$V_{G7} = -11.59V = V_D$$

$$R_{REF} \Rightarrow \frac{V_{DD} - V_D}{I_{ref}} \Rightarrow \frac{15 + 11.9}{5mA}$$

$$\Rightarrow \frac{26.6K}{5} \Rightarrow 5.3K\Omega$$

When used R_{Ref} value as $5.3K\Omega$ in LT spice the value for ~~I_{Ref}~~ I_{D4} ~~of~~ came as ~~4.92 mA 4.9~~ mA which is close or more or less equal to 5mA. 5.015

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PAGE